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TITLE: BEAUTYTRY: A VIRTUAL COSMETIC TRY-ON MOBILE APPLICATION

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ABSTRACT

Selecting suitable cosmetic products has become a common challenge among consumers especially when purchasing makeup from cosmetics retail stores. Many customers rely on physical cosmetic testers to evaluate product shades before making a purchase. However, the shared use of cosmetic testers raises hygiene concerns, as they may be contaminated with bacteria and viruses, increasing the risk of skin irritation and infection. In addition, some customers feel hesitant to use testers due to cleanliness issues. While others may purchase unsuitable products because they cannot confidently determine the right shade.

To address this issue, our group proposes BeautyTry: a Virtual Cosmetic Try-On Mobile Application that enables users to virtually try on cosmetic products using our prototype product camera. The application allows users to preview different shades of lipsticks products on their faces in real time without direct physical contact like testers. It also provides product information and shades recommendations based on the users facial features and skin tone. The expected outcome of this project is to provide a safer, more hygienic and convenient shopping experience while helping customers make more confident purchasing decisions. Furthermore, the application can reduce dependence on physical cosmetic testers, improve customer satisfaction and support digital transformation in the beauty retail industry.

CHAPTER 1 (INTRODUCTION)

1.1 INTRODUCTION

The cosmetics business has had trouble providing its customers with an opportunity to purchase a lipstick shade that will complement their skin color and lip shape. Retailers like Munikasumi Beauty House have been challenged with how to help their customers find the perfect shade of lipstick. Many traditional methods for selecting lipstick products involve the use of product testing samples by customers. Not only does this raise many sanitary issues, it may also pose some serious health risks for those who are sampling. In addition to these problems, when consumers try out different shades of lipstick, they are usually able to find one that is completely incorrect for them. This creates a situation where the customer is unhappy with the shade of lipstick they chose and has wasted money buying it. The result is that the customer loses faith in the retailer and becomes less loyal to the brand.

With the above in mind, this research calls for the development of the BeautyTry: A Virtual Cosmetic Try-On Mobile Application an innovative smart device combining HD camera, full-spectrum LED light, and AI technologies to determine lip color and texture accurately. This smart device deviates from the traditional lipstick shade testing method through emphasizing on non-contact technique which offers more safety and convenience to consumers. In addition, the interactive display will be helpful in offering real-time recommendations and virtual try-on of lipsticks shades, thus enabling customers to make the best decision on the desired product.

This research, accordingly seeks to explore the success of using the BeautyTry: Virtual Cosmetic Try-On Mobile Application in enhancing the consumer's experience, making fewer mistakes in the process of selecting products and building the brands credibility in cosmetic retailers like Munikasumi Beauty House. Solving the problem of proper choice of lipstick shades, removing the issues of hygiene during product testing and preventing consumers from buying wrong lipstick shades, this research aims at developing personalized beauty technologies and responding to the increasing needs for smart and sustainable retailing solutions.

1.2 PROJECT BACKGROUND

The beauty and cosmetics industry has grown rapidly as consumers increasingly rely on online platforms to purchase beauty products. However, one of the main challenges faced by customers when buying cosmetics online is the difficulty in visualizing how a product, especially makeup shades such as lipstick and foundation will look on their own face. This issue may cause uncertainty, dissatisfaction and product returns due to unsuitable choices.

To overcome this problem, BeautyTry is developed as a virtual beauty try-on application that utilizes Augmented Reality (AR) technology to allow users to try cosmetic products digitally before making a purchase. The application provides a more interactive and convenient shopping experience by helping users visualize suitable products based on their appearance.

This project aims to improve customer confidence in selecting beauty products while supporting cosmetic retailers in providing a modern and engaging digital shopping experience by helping customers make more confident purchasing decisions while reducing dependence on shared cosmetic testers. Additionally, BeautyTry supports cosmetics retailers by providing a more engaging and modern approach to showcase their products to customers.

1.3 PROBLEM STATEMENT

1. Difficulty choosing the right lip matte shade

- Many customers find it difficult to choose the most suitable lip matte shade when shopping online because they cannot test the product before purchasing.

2. Risk of purchasing the wrong product

- Customers may receive a lip matte shade that does not match their skin tone or personal preference, leading to dissatisfaction and unnecessary spending.

3. Hygiene concerns when testing lip matte products

- Many customers are concerned about hygiene when using lip matte testers in physical stores because the same product may be used by multiple people. This increases the risk of bacteria transfer and discourages customers from trying cosmetic products.

1.4 RESEARCH OBJECTIVE

1. To develop a virtual lip matte try-on application.
2. To provide a hygienic way to try lip matte products.
3. To reduce the risk of purchasing the wrong lip matte shade

1.5 RESEARCH QUESTIONS

1. How can a virtual lip matte try-on application help customers choose the right shade?
2. How can the application provide a hygienic way to try lip matte products?
3. How can the application reduce the risk of purchasing the wrong lip matte shade?

1.6 PROJECT SCOPE

This research focuses on the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application, a prototype application designed to help users identify the most suitable lipstick shades based on their facial features and skin tone using artificial intelligence. The study emphasizes the use of AI-based image analysis to provide personalized lipstick recommendations through an easy-to-use digital platform.

The scope of this research is limited to the development and testing of the prototype application. The system is designed to analyze users' facial images and recommend lipstick shades from a predefined database. Although the study primarily targets individuals interested in cosmetics and beauty products, the findings are intended to demonstrate the potential application of AI technology in the beauty industry, particularly in enhancing customer shopping experiences and personalized beauty recommendations.

1.7 IMPORTANCE OF STUDY

The significance of this study lies in its contribution to both the beauty industry and the advancement of artificial intelligence technology. BeautyTry: Virtual Cosmetic Try-On Mobile Application provides users with a convenient and efficient way to discover lipstick shades that complement their skin tone and facial appearance, reducing the uncertainty often associated with purchasing cosmetic products.

Furthermore, this study demonstrates how artificial intelligence can be integrated into the cosmetics industry to improve customer satisfaction and support informed purchasing decisions. The project also serves as an educational platform by showcasing the practical implementation of AI, image processing, and software development in solving real-world problems. By promoting personalized beauty recommendations through digital technology, this study contributes to innovation and the digital transformation of the beauty industry.

1.8 OPERATIONAL FINDINGS

Operational findings from this research show that the virtual cosmetic try-on mobile application performs perfectly well in ensuring a smart and clean cosmetic product retail environment. Integration of HD smart cameras along with the full spectrum LED lights helps the device in capturing lip color and texture data accurately, which is then processed by the AI system for recommending suitable lipstick shades. In the process of testing, the system proved its capability to decrease customer doubts through real-time results with high accuracy in terms of matching. Moreover, the system also helped in engaging customers through the virtual try-on feature, as it allowed them to see the effect of various lipstick shades on their lips without applying any product physically. This also helped in overcoming the hygiene issue associated with the traditional methods.

1.9 EXPECTED FINDINGS

This project is expected to help customers choose the most suitable lip matte shade through an AI-powered virtual try-on feature. The application also provides a hygienic alternative to physical cosmetic testers, allowing customers to try different shades safely and confidently before making a purchase while reducing the risk of buying the wrong product.

Otherwise, this project is expected to improve the overall online cosmetic shopping experience by increasing customers' confidence in selecting lip matte products. It may also benefit cosmetic retailers by enhancing customer satisfaction, reducing incorrect purchases, and encouraging more informed purchasing decisions.

1.10 SUMMARY

The above chapter has highlighted the basis for this study, which includes the background, problem statement, objectives, and scope of the study. This study is significant since it addresses actual problems being faced by cosmetics retailers and introduces innovation by incorporating the use of artificial intelligence in customized beauty solutions. Operational findings indicate that there is potential in the BeautyTry: A Virtual Cosmetic Try-On Mobile Application to enhance hygiene, accuracy and satisfaction when choosing products. In the following chapter, there will be a review of relevant literature in order to provide theoretical justification for this proposed solution.

CHAPTER 2 (LITERATURE RESEARCH)

2.1 INTRODUCTION

In recent times, there have been notable developments in the field of cosmetics due to the rising need for customizing personal beauty products in accordance with consumer taste and retail practices. One of the key issues that remain constant when it comes to consumers and retailers is the choice of an appropriate lipstick color in order to match the natural lip color and feel. One of the drawbacks associated with conventional testing methods is that they entail the direct application of testers to the lips.

Retailers such stores like Munikasumi Beauty House have faced difficulties in maintaining customer satisfaction because of these shortcomings, as consumers often buy lipstick colors which may not be suitable for them, resulting in disappointment, unnecessary expense, and loss of customers' trust in the store. This problem shows the urgent need for an efficient solution that could give consumers an opportunity to receive a reliable and personalized recommendation on the best lipstick color that suits them.

Therefore, this research suggests creating the Beautytry: A Virtual Cosmetic Try-on Mobile Application, which will be an innovative device and mobile application that will have high accuracy in the analysis of the lip color and texture through HD cameras, full spectrum LED lights and sophisticated AI algorithms. One of the main focuses of the system is on the non-contact and hygienic nature of its operation, as well as the virtual try-on function for consumers to see lipstick shades before buying them. This research aims to investigate the efficiency of this invention.

2.2 PREVIOUS STUDY

Virtual Cosmetic Applications

MELAKA: A survey conducted among university students revealed that virtual cosmetic applications significantly improved confidence in product selection, especially for lipstick and foundation shades. Respondents reported that AR-based try-on systems reduced uncertainty and minimized financial waste. “Virtual try-on technology allows consumers to visualize products before purchase, reducing dissatisfaction,” the researchers concluded. (Nurliyana Abas et al., 2026)

PETALING JAYA: A dermatology clinic reported a rise in cases of allergic contact dermatitis among women who had recently used cosmetic testers in shopping malls. The Malaysian Dermatology Association confirmed that several patients developed rashes and swelling after applying lip products directly from shared testers. “Our findings show that improper hygiene practices in cosmetic retail outlets can trigger allergic reactions and worsen skin conditions,” said the lead dermatologist. The report recommended the adoption of virtual try-on applications to reduce direct contact with testers and improve consumer safety. (The Sun Malaysia, 13.9.2025)

SHAH ALAM: A university research team conducted a study on microbial contamination in cosmetic testers across five retail outlets. Laboratory analysis revealed the presence of *Staphylococcus aureus*, *E. coli*, and *Pseudomonas aeruginosa* in lipstick and foundation testers. “The results confirm that shared cosmetic testers are a potential medium for spreading infectious diseases,” the researchers stated. The study concluded that safer alternatives such as AI-based virtual try-on systems should be implemented to protect consumers from health risks. (Kim et al., 2025)

JOHOR BAHRU: A consumer rights group highlighted the issue of wrong shade purchases among cosmetic buyers, noting that poor store lighting and inaccurate packaging references often mislead customers. The group reported that many consumers wasted money on unsuitable products or were forced to mix shades to achieve the desired look. “This problem reduces customer confidence and contributes to product wastage,” the report stated. The group recommended the integration of AR and AI technologies in retail outlets to provide real-time shade matching and minimize financial losses. (Carlsson, 2026)

Health Risks of Testers

PETALING JAYA: Dermatology experts warned that cosmetic testers can act as a medium for spreading bacteria, fungi, and viruses such as HSV-1 (Herpes Simplex Virus). A study conducted by the Malaysian Dermatology Association found that improper hygiene practices in retail outlets triggered allergic reactions and worsened skin conditions among consumers. “Our findings show that shared testers are unsafe and can cause serious health complications,” said the lead dermatologist. The report recommended virtual cosmetic applications as a safer alternative. (The Sun Malaysia, 13.9.2025)

SEOUL: A laboratory study conducted by Kim et al. (2025) analyzed microbial contamination in cosmetic testers across multiple retail outlets. The results revealed the presence of *Staphylococcus epidermidis*, *Pseudomonas aeruginosa*, *Streptococcus spp.*, and *E. coli* in lip and foundation testers. “The findings confirm that cosmetic testers are a potential medium for spreading infectious diseases,” the researchers concluded. The study emphasized the importance of adopting AI-based virtual try-on systems to reduce direct contact and protect consumer health.

BRAZIL: Fagundes et al. (2025) carried out a microbiological evaluation of cosmetic testers in retail environments and discovered contamination with *Staphylococcus aureus*, *E. coli*, and *Pseudomonas aeruginosa*. The study highlighted that shared testers pose biosafety risks and can lead to outbreaks of skin infections. Researchers strongly recommended the implementation of non-contact alternatives such as sensor-based or virtual try-on technologies to ensure consumer safety.

Wrong Shade Risks

MELAKA: A survey conducted among university students revealed that many cosmetic buyers struggled with wrong shade purchases due to inaccurate packaging references. Respondents admitted that they often wasted money on unsuitable lipsticks and foundations, which reduced their confidence in cosmetic shopping. “Shade mismatch leads to dissatisfaction and unnecessary spending,” the researchers concluded. The study recommended the adoption of virtual try-on applications to minimize errors. (Carlsson, 2026)

NEW YORK: An industry report by Arbelle (2026) found that foundation shade mismatch is a major driver of product returns in the cosmetics sector. The report highlighted that undertones were poorly represented in digital catalogs, leading to high dissatisfaction among consumers. “Shade mismatch is a critical issue that weakens brand loyalty and increases financial losses,” the report stated. The findings emphasized the importance of advanced shade-matching technologies.

LONDON: The Safety in Beauty Foundation (2026) reported that counterfeit cosmetics worsen shade mismatch risks, as fake products often mislabel shade codes and contain harmful ingredients. Several consumers experienced irritation and allergic reactions after using counterfeit lipsticks purchased online. “Counterfeit cosmetics mislead customers about shade accuracy and pose health risks,” the foundation warned. The study recommended stricter regulation and verified digital shade-matching tools.

TOKYO: A consumer study revealed that poor store lighting and lack of tester alternatives caused frequent wrong shade purchases among Japanese cosmetic buyers. Customers reported frustration at having to mix products or discard unsuitable shades. “This problem contributes to product wastage and reduces consumer trust,” the study noted. Researchers suggested integrating AR and AI technologies in retail outlets to provide real-time shade matching and improve customer satisfaction. (Personal Care Products Council, 2024)

2.3 SUMMARY

The review of literature clearly indicates that conventional practices in cosmetic retail, such as the use of physical testers and reliance on packaging references, are increasingly ineffective and problematic. Manual testing methods not only expose consumers to health risks through microbial contamination and viral transmission but also lead to dissatisfaction due to frequent shade mismatches. These shortcomings result in unnecessary financial losses, reduced customer confidence, and weakened brand loyalty. In contrast, previous studies have consistently demonstrated that virtual cosmetic applications integrating AI and AR technologies provide safer, more efficient, and user-friendly solutions. By enabling real-time shade visualization, personalized recommendations, and non-contact testing, these systems address hygiene concerns while simultaneously improving consumer satisfaction and reducing product wastage. Therefore, the findings strongly support the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application, which is expected to enhance the effectiveness of cosmetic retail operations, minimize risks, and align with modern digital and sustainable practices.

CHAPTER 3 (METHODOLOGY/PROJECT DESIGN)

3.1 INTRODUCTION

This chapter elaborate more about methodology and design for our project BeautyTry : A Virtual Cosmetic Try-On Mobile Application. This methodology meets the needs and expectations of users in selecting suitable lip matte products. This application aims to provide a more convenient and interactive experience by allowing users to visualize different lip matte shades before making a purchase decision.

To ensure the application is developed effectively. This systematic methodology is implemented throughout the project development process. Methodology consists of four main stages which is problem identification, data collection, design development and design evaluation. Problem identification is conducted to understand the challenges faced by users, while data collection focuses on gathering relevant information and user requirements. The design development stage involves creating the application based on the identified needs, followed by design evaluation to determine the usability and effectiveness of the application.

Through this approach, the project aims to produce an application that can improve user confidence in choosing lip matte products while providing a safer and more efficient alternative to traditional cosmetic testing methods.

3.2 STUDY DESIGN

This design study focuses on the development of a BeautyTry: A Virtual Cosmetic Try-On Mobile Application that aims to assist users in selecting suitable lipstick shades based on their skin tone and facial features through a digital platform. The system design incorporates Artificial Intelligence (AI) technology by using a camera, facial scanning system, and image analysis to detect the user's facial features, especially the lip area, accurately.

To support the implementation of this application during an event or product demonstration, the system is designed with several main components, including a face detection camera, proper lighting equipment to illuminate the user's face, and a tablet screen that displays the application interface and real-time lipstick shade simulation results. The lighting system helps the camera capture clearer facial images, while the facial scanning technology enables the application to analyze facial characteristics and apply suitable lipstick shades digitally.

Furthermore, the design of this application considers important aspects such as user-friendly interaction, accurate colour detection, and an enhanced virtual try-on experience. Through the integration of hardware and software components, users can explore and select lipstick shades confidently without the need for physical product testers. This approach helps to improve hygiene practices, reduce the risk of product contamination, and provide a more convenient and engaging shopping experience within the beauty industry.

3.2.1 SAMPLING

The sampling method used in this study was purposive sampling, whereby respondents were selected based on specific criteria that aligned with the objectives of the research. A total of 30 respondents participated in the questionnaire survey. The respondents consisted of individuals who had experience purchasing or using cosmetic products, particularly lip products such as lipstick, lip tint, and lip matte. They were chosen because they represent the target users of the proposed system, BeautyTry: A Virtual Cosmetic Try-On Mobile Application

The questionnaire focused on respondents' experiences and challenges when selecting lipstick shades, including issues such as inaccurate colour representation on product packaging, unsuitable lighting conditions in cosmetic stores, the unavailability of product testers, and concerns regarding the hygiene of shared testers. In addition, respondents were asked to provide their opinions on whether an Artificial Intelligence (AI), based on system could help reduce colour selection errors, minimise product wastage and improve the overall shopping experience.

The data collected from the 30 respondents provided valuable insights into users needs, preferences, and acceptance of AI technology in cosmetic shade selection. These findings served as an important reference for the design and development of the BeautyTry: A Virtual Cosmetic Try-On Mobile Application, ensuring that the proposed system meets user requirements and addresses the challenges identified through the survey.

3.2.2 DATA COLLECTION METHODS

The first method involved distributing questionnaires to collect information from respondents regarding their experiences in selecting the appropriate lip matte shade, the risk of purchasing an unsuitable product, and hygiene concerns associated with testing lip matte products. The structured format of the questionnaire allowed the data to be collected consistently, making the responses easier to analyse and providing valuable insights into customer preferences and experiences.

In addition to the questionnaires, direct observation was carried out at Muni Kasumi Beauty House. During the observation, it was found that not all lip matte products were provided with tester units. As a result, customers often faced difficulties in identifying their desired shade before making a purchase, which could reduce customer satisfaction and increase the likelihood of buying an unsuitable product. Furthermore, the availability of tester products created additional challenges for the retailer, as the tester stock was frequently depleted and required replacement. Hygiene issues were also observed due to multiple customers using the same tester products.

The findings from the direct observation were consistent with the questionnaire results, strengthening the conclusion that customers experience challenges in selecting the correct lip matte shade, while retailers at Muni Kasumi Beauty House face significant issues related to tester availability, stock management, and hygiene.

3.2.3 DATA ANALYSIS METHODS

The data collected from the questionnaire were analyzed using quantitative methods. Descriptive statistics, including frequencies and percentages, were used to analyze the respondents' opinions and experiences regarding lipstick shade selection and the need for the proposed BeautyTry application. The results were presented in the form of charts and graphs to facilitate data interpretation.

Based on the analysis, 90.5% of the respondents identified lipstick as the most difficult cosmetic product to purchase online. In addition, 50% of the respondents stated that they had purchased a cosmetic product whose actual shade differed from their expectations after application, while 21.4% reported that this issue occurred frequently. These findings indicate that many consumers experience difficulties in selecting the appropriate lipstick shade.

The findings also revealed that 64.3% of the respondents believed that different lighting conditions in stores were the main factor affecting the appearance of lipstick shades, followed by 23.8% who identified mobile phone screen lighting as another contributing factor. Furthermore, 45.2% of the respondents stated that choosing the wrong lipstick shade resulted in wasted money, while 40.5% reported that they had to purchase another product because the selected shade was unsuitable.

Regarding hygiene concerns, 69% of the respondents strongly agreed and 31% agreed that sharing cosmetic testers among customers could lead to hygiene issues. These findings suggest that consumers prefer a safer and more convenient alternative for trying lipstick shades before making a purchase.

Overall, the data analysis indicates that consumers face significant challenges in selecting suitable lipstick shades and require a more accurate and convenient solution. Therefore, the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application is proposed to assist users in identifying suitable lipstick shades based on their skin tone using Artificial Intelligence (AI). This application is expected to reduce purchasing errors, improve user confidence, and provide a more hygienic alternative to conventional cosmetic testers.

3.3 SUMMARY

This chapter has outlined the methodology and project design adopted in the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application. The research process began with problem identification, followed by purposive sampling of 30 respondents who represented the target users. Data was collected through questionnaires and secondary sources such as news reports and analyzed using both quantitative and qualitative methods. The findings confirmed that users face significant challenges in shade selection due to inaccurate packaging references, unsuitable store lighting and hygiene concerns with shared testers. Based on these insights, the BeautyTry system was designed to integrate AI technology, facial scanning and real-time shade simulation, providing a user-centered, hygienic and efficient solution. The structured methodology ensured that the application meets user needs while offering a safer and more interactive alternative to traditional cosmetic testing.

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APPENDIX

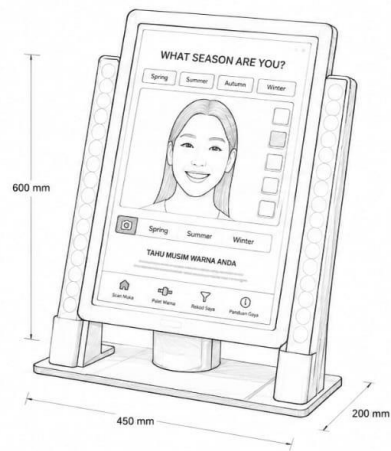
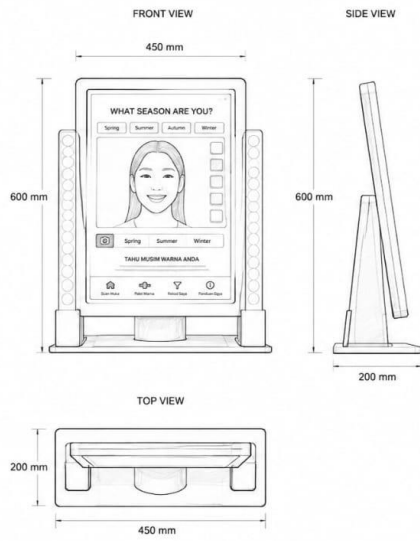
1. Project Cost Estimation

COMPONENT	QUANTITY	UNIT PRICE (RM)	TOTAL (RM)
IPad 11 th Generation 11inch (RAM 4GB+)	1	1,371.00	1,371.00
LED Fill Light Bar	2 pieces	22.00	44.00
Adjustable Tablet Holder	1	18.00	18.00
Acrylic Sheet 5mm	1	36.00	36.00
MDF Board / Acrylic Base	1	31.00	31.00
Hollow Steel Frame	4 pieces	18.00	72.00
Skru, Nut and Bolt	1 set	12.00	12.00
Epoxy / Acrylic Glue	1	14.00	14.00
Extension Plug	1	21.00	21.00
Paint / Vinyl Sticker	2	27.00	54.00

Total Cost: RM1,673.00 +-

2. Product Sketch

Sketch The Design



EXPLANATION : BeautyTry : A Virtual Cosmetic Try-On Mobile Application

The prototype design of BeautyTry: A Virtual Cosmetic Try-On Mobile Application was carefully planned to provide users with an interactive, hygienic, and convenient cosmetic testing experience. The prototype consists of a tablet mounted on a stable stand, integrated with a front-facing camera and dual LED light bars. The design enables users to virtually try different lipstick shades in real time without the need for physical cosmetic testers.

From the **front view**, the prototype features a tablet that serves as the main user interface. The application displays a live camera feed and provides several lipstick shade options for users to select. Once a shade is chosen, the application uses artificial intelligence (AI) facial detection technology to accurately identify the user's lips and instantly apply the selected lipstick colour on the screen. The interface also includes navigation buttons that allow users to switch between lipstick shades, compare colours, and save their preferred selection.

From the side view, the tablet is positioned at an ergonomic viewing angle to ensure user comfort during the virtual try-on session. Behind the tablet, a support frame securely holds the device in place, while internal cable management keeps the overall appearance neat and organised. Two LED light bars are installed on both sides of the tablet to provide consistent illumination, improving facial detection accuracy and ensuring realistic colour rendering under different lighting conditions.

In terms of **hardware components**, the prototype includes an Android tablet, a built-in or external HD camera, dual LED fill lights, a tablet holder, and a sturdy base constructed from acrylic or MDF material. The tablet runs the BeautyTry mobile application, while the AI-based facial recognition system processes the user's face and overlays virtual lipstick shades in real time. The LED lighting enhances image quality, allowing the application to produce more precise and natural-looking cosmetic simulations.

The **overall structure** is designed to be compact, portable, and user-friendly, making it suitable for use in cosmetic stores, shopping malls, beauty exhibitions, and promotional events. By replacing traditional cosmetic testers with virtual technology, the BeautyTry prototype helps improve hygiene, reduces product wastage, and enables customers to make more confident purchasing decisions through an engaging digital experience.